Workshop-2

Go to your Computer laboratory and try to identify the type of computer network followed there. Seek help from your teacher.

14.4. Wired Technologies

The means through which data is transferred from one place to another is called transmission or communication media. There are two categories of transmission media used in computer communications.

- Wired/Bounded/Guided media
- Wireless/Unbounded/Unguided media

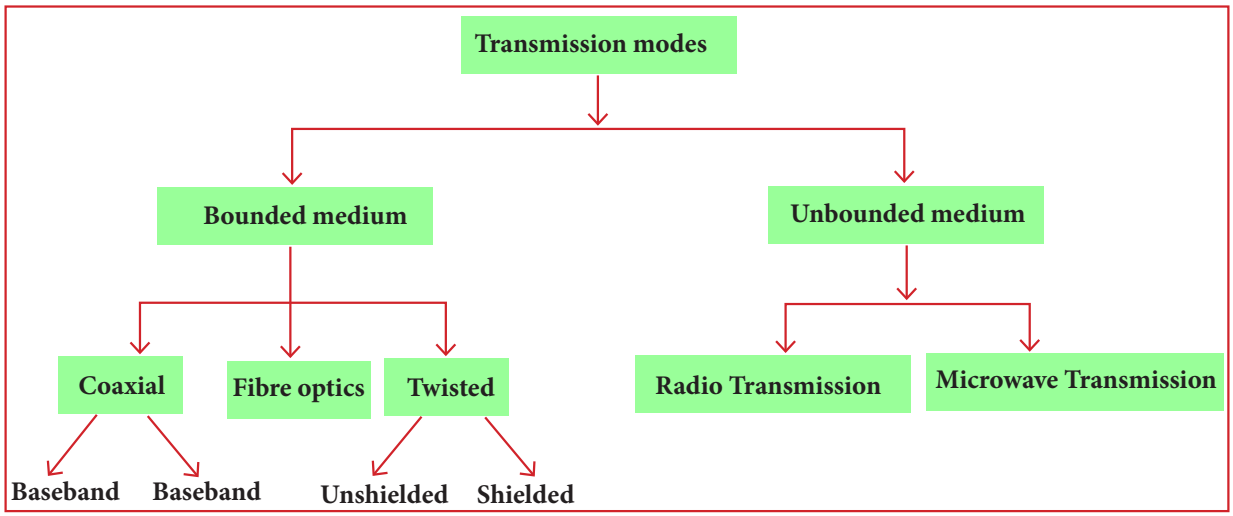

Figure 14.8 Wired Technologies

Bounded/Guided Media

There are three common types of bounded media in the data transmission. They are

- Twisted-Pair Cable,
- Coaxial Cable, and
- Fibre-Optic Cable.

Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Optical fibre is a cable that accepts and transports signals in the form of light.

14.4.1. Twisted Pair Cable

A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together. One of these wires is used to carry signals to the receiver, and the other is used only as ground reference. In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.

Twisted Pair is of two types:

- Unshielded Twisted Pair (UTP)
- Shielded Twisted Pair (STP)

Unshielded Twisted Pair Cable